## Literature Review

Financial fraud detection has evolved rapidly with machine learning (ML) techniques, shifting from rule-based reactive systems to intelligent, proactive models capable of real-time prevention. A comprehensive systematic literature review by Hernandez Aros et al. [1] analyzed 104 articles published between 2012 and 2023 using adapted PRISMA and Kitchenham protocols. The study employed a two-stage search strategy across Scopus, IEEE Xplore, Taylor & Francis, SAGE, and ScienceDirect, applying strict inclusion/exclusion criteria (e.g., full-text ML-focused studies on financial fraud, minimum 5 citations, 2012–2023 timeframe). After duplicate removal and quality assessment via a 7-question framework, 86 experimental papers were synthesized alongside 18 secondary reviews.

Key findings from [1] reveal supervised learning as the dominant paradigm (56.73% of studies), with Random Forest (RF) emerging as the most frequently applied model (34 mentions), followed by Logistic Regression (32), Support Vector Machine (29), and Decision Tree (29). Deep learning approaches, including Long Short-Term Memory (LSTM) networks, were employed in multiple studies for sequence-based fraud detection, with LSTM specifically cited 7 times for its strength in processing temporal transaction patterns. The review explicitly notes: "LSTM model is a recurrent neural network used for sequence processing, especially for tasks concerning natural language processing," and highlights its role within supervised/deep learning hybrids (2.88% of models). Datasets were overwhelmingly real-world (over 93% usage, synthetic data <7%), with the most popular being the Credit Card Fraud Detection dataset (284,807 transactions, 0.172% fraud rate) and others drawn from stock exchanges in China, Canada, the United States, Taiwan, and Tehran. Performance was evaluated primarily via accuracy, precision, recall, F1-score, and AUC derived from confusion matrices. Country contributions placed China first (18 papers) and India second (13 papers), underscoring India's prominence in credit card and loan fraud research. Major challenges identified include severe class imbalance, privacy concerns, high misclassification costs, and the persistent gap between post-authorization detection and true pre-authorization prevention. The authors recommend exploring advanced deep-learning and real-time anomaly detection techniques to address emerging fraud vectors such as velocity-based and behavioral anomalies.

Building directly on these insights, Mukhamadiyev et al. [2] propose an intelligent LSTM-based framework optimized with the Artificial Bee Colony (ABC) algorithm for suspicious transaction detection in payment data management systems. The methodology processes sequential transaction data through a full LSTM architecture featuring input, forget, output, and cell gates, with binary cross-entropy loss and backpropagation through time. To overcome manual hyperparameter tuning limitations, ABC optimization automatically searches for optimal hidden units (10–200) and learning rates (0.00001–0.1) via population-based neighbor generation and fitness evaluation on the loss function. The experimental dataset, sourced from real partner financial organizations, includes features such as transaction amount, timestamps (broken into hour/day components for temporal analysis), payment method, location, IP address, and merchant ID — directly enabling velocity tracking and behavioral deviation detection. Pre-processing steps covered normalization, median imputation for missing values, Z-score outlier handling, and categorical encoding.

Comparative results in [2] demonstrate clear superiority of the hybrid LSTM+ABC model over traditional baselines: 95% accuracy, 92% precision, 97% recall, and 94% F1-score, outperforming standalone LSTM (93% accuracy, 92% F1), XGBoost (91.65% accuracy), Random Forest (90.6% accuracy), SVM (93.75% accuracy), Logistic Regression, and KNN. The study emphasizes real-time applicability, stating that the approach "creates significant opportunities for effective risk management in real time and improving payment security." It particularly excels at capturing nonlinear temporal dependencies and complex fraudulent patterns in imbalanced data (fraud rate ≈1.5%), making it highly suitable for pre-authorization scenarios where decisions must occur before payment processing.

These two works provide a strong foundation for Project ARGUS. The broad taxonomy and recommendations in [1] validate the need for hybrid ensembles that combine supervised models (such as our XGBoost and Isolation Forest) with sequence-aware deep learning (LSTM for behavioral velocity patterns across 1-min/5-min/1-hr/24-hr windows). The targeted LSTM+ABC implementation in [2] directly mirrors ARGUS's multi-model architecture and 45+ feature engineering pipeline, confirming that optimized LSTM achieves higher recall on temporal anomalies (SIM swap timing, mule account rapid flows, impossible travel) while maintaining sub-20 ms latency targets. By addressing the exact gaps highlighted in [1] — class imbalance, real-time shift from reactive to proactive detection, and India's growing research focus — ARGUS delivers explainable pre-authorization decisions, graph-based ring detection, and RBI/NPCI compliance. The integration of LSTM for sequence modeling (as recommended in both studies) will further elevate detection accuracy beyond 98.7% while minimizing false positives, proving the technical viability of our prototype.

## References


[1] L. Hernandez Aros, L. X. Bustamante Molano, F. Gutierrez-Portela, J. J. Moreno Hernandez, and M. S. Rodríguez Barrero, "Financial fraud detection through the application of machine learning techniques: a literature review," Humanities Social Sci. Commun., vol. 11, no. 1, p. 1130, 2024, doi: 10.1057/s41599-024-03606-0.

[2] A. Mukhamadiyev, F. Nazarov, S. Yarmatov, and J. Cho, "Intelligent algorithms for the detection of suspicious transactions in payment data management systems based on LSTM neural networks," Sensors, vol. 25, no. 21, p. 6683, 2025, doi: 10.3390/s25216683.